

Morphogenesis and morphometry of brain folding patterns across species

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eLife Assessment

This **important** study presents a cross-species and cross-disciplinary analysis of cortical folding. The authors use a combination of physical gel models, computational simulations, and morphometric analysis, extending prior work in human brain development to macaques and ferrets. The findings support the hypothesis that mechanical forces driven by differential growth can account for major aspects of gyrification. The evidence presented, though limited in certain species-specific and parametric details, is overall strong and **convincingly** supports the central claims; the findings will be of broad interest in developmental neuroscience.

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Abstract

Evolutionary adaptations associated with the formation of a folded cortex in many mammalian brains are thought to be a critical specialization associated with higher cognitive function. The dramatic surface expansion and highly convoluted folding of the cortex during early development is a theme with variations that suggest the need for a comparative study of cortical gyrification. Here, we use a combination of physical experiments using gels, computational morphogenesis, and geometric morphometrics to study the folding of brains across different species. Starting with magnetic resonance images of brains of a newborn ferret, a fetal macaque, and a fetal human, we construct two-layer physical gel brain models that swell superficially in a solvent, leading to folding patterns similar to those seen *in vivo*. We then adopt a three-dimensional continuum model based on differential growth to simulate cortical folding *in silico*. Finally, we deploy a comparative morphometric analysis of the *in vivo*, *in vitro*, and *in silico* surface buckling patterns across species. Our study shows that a simple mechanical instability driven by differential growth suffices to explain cortical folding and suggests that variations in the tangential growth and different initial geometries are sufficient to explain the differences in cortical folding across species.

Introduction

Although not all brains are folded, in many mammals, the folded cerebral cortex is known to be critically important for brain cognitive performance and highly dependent on the hierarchical structure of its morphology, cytoarchitecture, and connectivity [1, 2, 3]. Brain function is thus related both to the topological structure of neural networks [4], as well as the geometry and morphology of the convoluted cortex [5], both of which serve to enable and constrain neuronal dynamics [3]. Across species, cortical morphologies show a large diversity, as shown in **Fig. 1(a)** [6, 7]. And within our own species and in model organisms such as the ferret used to study the genetic precursors of misfolding, cortical folding and misfolding are known to be markers of healthy and pathological neurodevelopment, disease and aging [8, 9] (see also (Fig. S1 [10, 11])). Thus a comparative study of cortical folding is essential for understanding brain morphogenesis and functionalization across evolution [3, 12], during development as well as in pathological situations associated with disease.

The development of cortical morphology involves the coordinated and localized expression of many genes that lead to the migration and differentiation of neural stem and progenitor cells [13, 10, 14]. All these biological processes cooperatively generate an expansion of the cortex relative to the underlying white matter and eventually drive cortical folding [15]. While a range of mechanisms have been proposed in the past for the processes leading to folding [16, 17, 18], over the past decade, theoretical and experimental evidence have converged on the primary determinant of folding as a mechanical instability associated with the formation of a localized crease or sulcus [19, 20] driven by differential growth, with iterations and variations that qualitatively explain the brain gyrification [21, 22]. However, combining this mechanistic model with a comparative perspective that aims to quantify the variability of folding patterns across species, or linking it to genetic perturbations that change the relative expansion of the cortex remain open questions. In a companion paper [11], we address the latter using the ferret as a model organism, while in the current study, we address the former question using a combination of physical experiments with gel swelling, numerical simulations of differential growth and geometric morphometrics to compare brain morphogenesis in the ferret, the macaque, and the human. The species were selected as they have very different brain sizes and folding patterns (**Fig. 1(b)**) and thus represent different branches in the evolutionary tree, Carnivora, Old World monkeys, and Hominoidea. Furthermore, in all three species, we have access to a time course of the development of the folds, as shown in **Fig. 1(c)**.

Experiments on swelling gel-brains

To mimic the mechanical basis for brain morphogenesis based on the differential growth of the cortex relative to the white matter, we used the swelling of physical gels that mimic the fetal brain developmental process during post-gestation stages. Previously, this principle has been demonstrated for *Homo sapiens* (human) brain development [22, 21]. To demonstrate that the same principle applies to other species, we constructed physical gels from the fetal brain MRI scans for *Macaca mulatta* (rhesus macaque) and *Mustela furo* (ferret). In brief, a two-layer PDMS gel is constructed from the 3D fetal brain MRI reconstruction and immersed in an organic solvent. Immersion causes solvent imbibition into the surface of the physical gel which swells leading to a compressive strain in the outer layers that causes the surface layer to form convolutions that resemble the folding patterns in the brain cortex layer. Time-lapse images of the gel model to mimic brain folding in *Macaca mulatta* (rhesus macaque) up to G110 are shown in **Fig. 2(a)**, while in **Fig. 2(b)**, we show the initial and final states of swelling to mimic different post-gestation stages corresponding to G85, G110 and G135. A visual inspection of the swollen gels constructed from different post-gestation stages showed qualitatively different folding patterns, indicating the sensitivity of the folds to the initial undulations present on the physical gel surfaces.

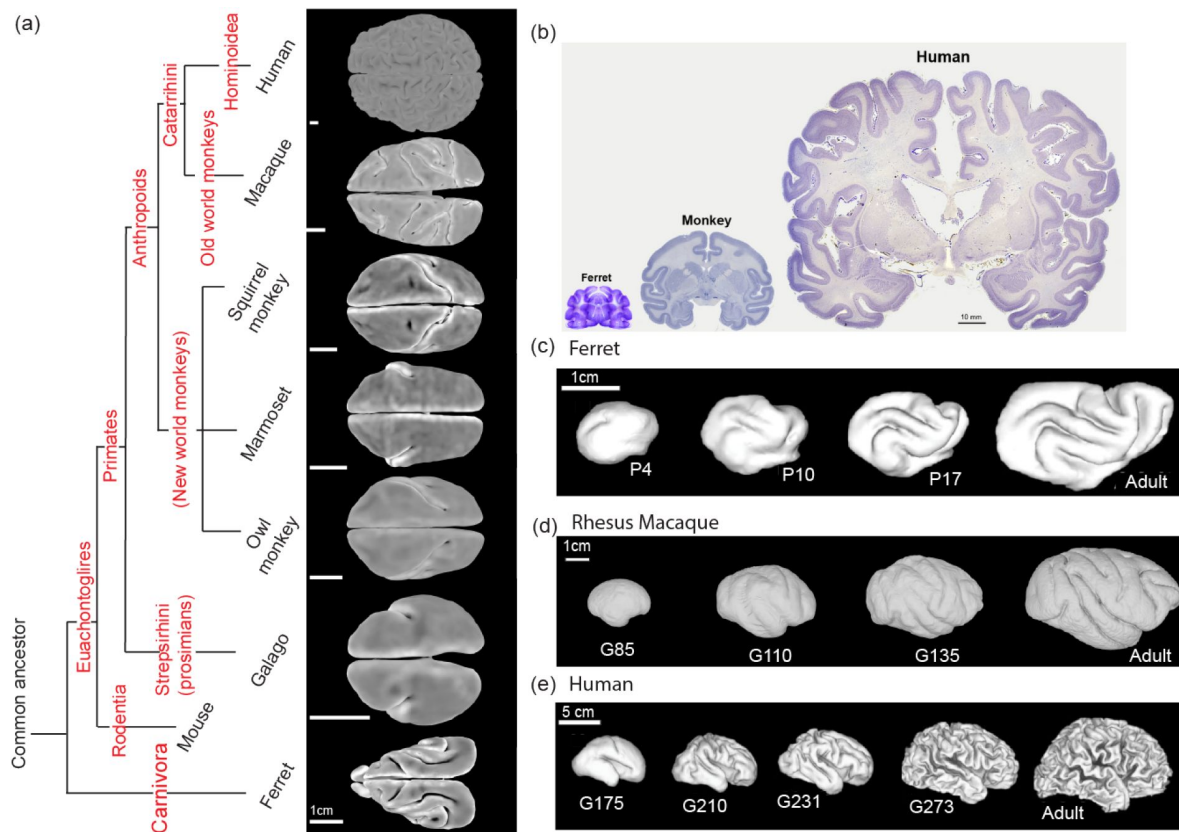


Fig. 1

The diversity of the cortical morphologies and developmental processes across species.

(a) Phylogenetic relationship of species. Adapted from [6, 7]. Typical real brain surfaces of ferret and primates are presented. Color represents mean curvature. Scale bars: 1 cm (estimated from [23]). (b) Stained sections of mature brain tissue from ferret, rhesus macaque, and human. Scale bar: 10 mm. Adapted from [24]. (c-e) 3D reconstruction of cortical surfaces of ferret, macaque, and human brains from fetal to adult. (c) Ferret: postnatal day 4, 10, 17 and adult maturation [25]. Scale bar: 1 cm; (d) Macaque: gestation day 85, 110, 135 [26], and adult maturation [27]. Scale bar: 1 cm; (e) Human: gestation day 175 (week 25), 210 (week 30), 231 (week 33), 273 (week 39), and adult maturation [25]. Scale bar: 5 cm.

In **Fig. 2(c)** and Movie S1, we show the results of similar physical gel experiments to mimic brain folding morphogenesis for *Homo sapiens* [22] and *Mustela furo* [11]; in each case the initial condition was based on 3d fetal brain MRIs and the final state was determined qualitatively using the overall volume of the brain relative to the initial state. No attempt was made to vary the swelling ratio of the surface as a function of location, although it is likely that in the different species this was not a constant. To quantitatively describe the folding patterns, the swollen gel surfaces were then scanned and reconstructed using X-ray Computed Tomography (Methods). Movie S2–S4 showed the fetal brain MRI scans and the reconstructed 3D swollen gel surfaces in juxtaposition for *H. sapiens*, *M. furo*, and *M. mulatta*. This paves the way for a quantitative comparison of the results of the physical experiments with those derived from a mechanical theory for brain morphogenesis and those from scans of macaque, ferret and human brains.

Simulations of growing brains

To test the capability of the mechanical model for brain morphogenesis based on differential growth [21, 28, 22] to explain patterns across species, we perform numerical simulations of the developing brains of ferret, macaque, and human modeled as soft tissues. Here we only consider the simplest homogeneous growth profile which is sufficient to capture the folding formation across different species; regional growth of the cortical layer based on real data from tracking the surface expansion of fetal brains [29, 30, 31] is a more sophisticated alternative that we do not adopt for reasons of simplicity. The initial brain models are reconstructed from 3D fetal brain MRI (Methods), and assumed to be composed of gray and white matter layers which are considered as hyperelastic materials with differential tangential growth ratio g . A multiplicative decomposition of deformation gradient gives $\mathbf{F} = \mathbf{A} \cdot \mathbf{G}$ with $\mathbf{A}$ the elastic part and $\mathbf{G} = \sqrt{g}\mathbf{I} + (1 - \sqrt{g})\mathbf{n} \otimes \mathbf{n}$ the growth tensor. We adopt a modestly compressible neo-Hookean material with strain energy density

$$W = \frac{\mu}{2} \left[J_A^{-2/3} \text{tr}(\mathbf{A}^T \cdot \mathbf{A} - 3) \right] + \frac{K}{2} (J_A - 1)^2, \quad (1)$$

where $\mathbf{F}$ is the deformation gradient, $J_A = \det \mathbf{A}$, μ is the initial shear modulus, and K is the bulk modulus. Considering modestly compressible material, we assume $K = 5\mu$.

We solve the final shapes of the growing tissues using a custom finite element method [22]. All the initial geometries of smooth fetal brains are obtained from open data sources (Methods, 3D model reconstruction), and the growth ratio distribution is assumed as a function of the initial location, including the distance to the cortical surface and the presumably non-growing regions. Other parameters, such as the thickness ratio and modulus ratio of gray and white matter, and the temporal changes of growth ratios are assumed (Methods, numerical simulations, Table 2). The brain models are discretized to tetrahedrons by Netgen. An explicit algorithm is adopted to minimize the total strain energy of the deforming tissues. We adopt a step-wise simulation strategy where the initial geometry of each presumed stress-free state is obtained from real MRI data of earlier-stage fetal brains, instead of using a continuous model where only the initial smooth brain geometry is input [22]. The simulated developmental processes of fetal brains are presented in Fig. 3 and Movie S1.

Morphometric analysis

To verify whether our simulation methods and physical gel models are sufficient to capture the developmental process of fetal brains and reproduce cortical patterns in adult brains, we compare the cortical surfaces of real, simulated, and gel brains across different species. Fig. 4(a) presents the hemispherical cortical surfaces of the real, simulated, and gel brains (denoted as $\mathcal{S}_1$, $\mathcal{S}_2$, and $\mathcal{S}_3$, respectively). Left and right symmetry and the comparison of whole brains are presented in Fig. S3 and S4, Movie S2–S4. Major sulci are extracted and highlighted by hand for further

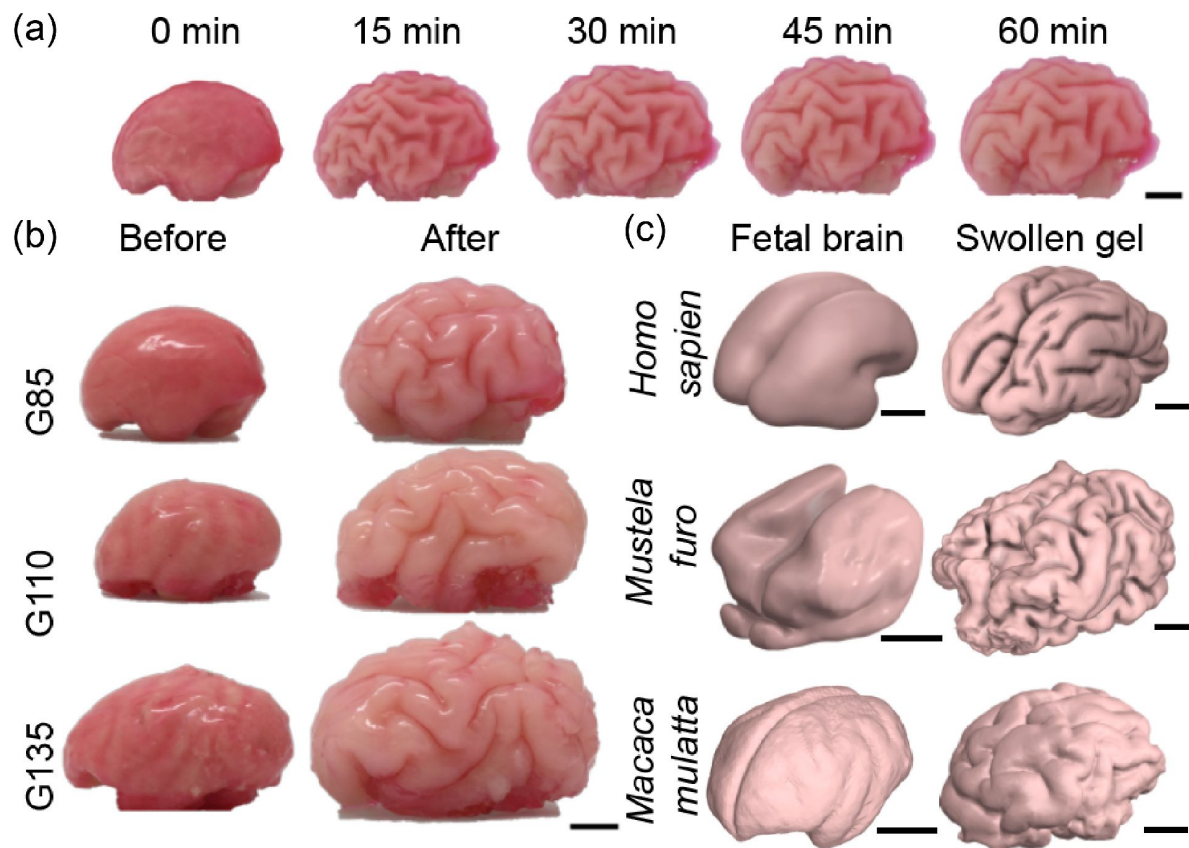


Fig. 2

Physical gel model that recapitulates the growth-driven morphogenesis mechanism across phylogeny and developmental stages.

(a) A time-lapse of the physical gel brain mimicking macaque brain development starting from G110. (b) Left views of three physical gels mimicking macaque post gestation day 85, day 110, and day 135 before and after hexane swelling. Scale bar: 1 cm. (c) Comparison of fetal/newborn brain scans and the reconstructed surfaces of swollen physical gels for various species. Scale bars: 1 cm.

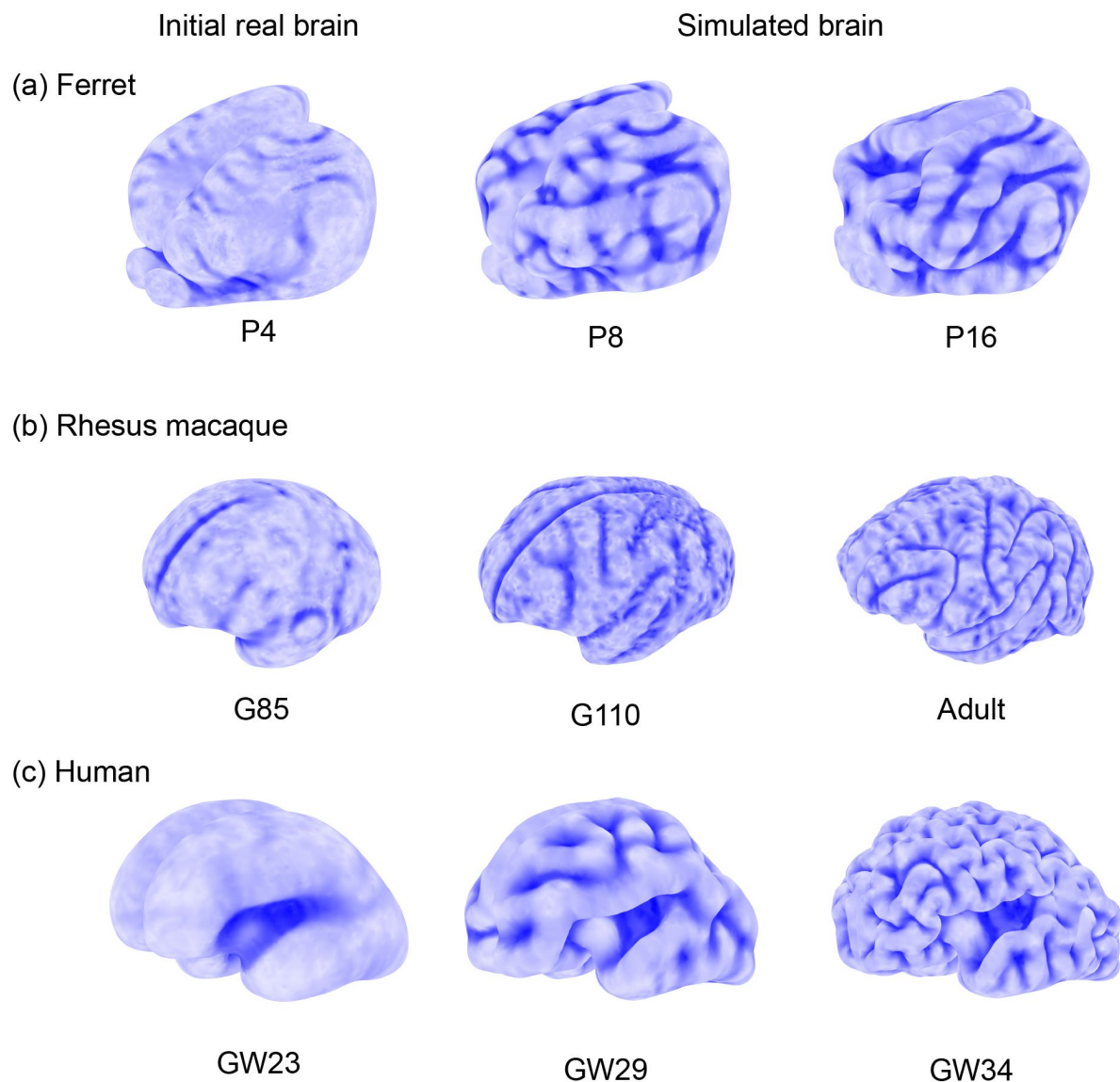


Fig. 3

Simulations of growing brains of

(a) ferret, (b) rhesus macaque, and (c) human. Starting from smooth fetal/newborn brains, simulations show different gyrification patterns across species. The brains are modeled as soft elastic solids with tangential growth in the gray matter (see *Simulations of growing brains* for details). Initial 3D geometries are taken from the reconstruction of MRI (see *Methods*, 3D model reconstruction). Mechanical parameters of growth ratio and cortical thickness are provided in **Table 2**. Color from dark to light blue represents shape index (as defined in **Eq. (2)**) from -1 to 1 .

alignment. To analyze the shape differences, we then adapted the parameterization methods in [32, 33, 34] to map the brain surfaces onto a common disk shape with the major sulci aligned using landmark-matching disk quasi-conformal parameterizations. Denote the three disk parameterization results as $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$, as shown in **Fig. 4(b)**. Multiple quantitative measures such as surface area, cortical thickness, curvature, and sulcal depth can be adopted to analyze curved surfaces and to compare different surfaces. Here, we use shape index (SI) and rescaled mean curvature $\tilde{H}$ to describe the similarity among real, simulated, and gel surfaces. Shape index is a dimensionless and scale-independent surface measure [35]. It can be calculated from the mean and Gaussian curvatures by

$$SI = \frac{2}{\pi} \arctan \left(\frac{H}{\sqrt{H^2 - K}} \right). \quad (2)$$

Shape index ranging within $[-1, 1]$ defines a continuous shape change from convex, saddle, to concave shapes. Fig. S2 shows nine categories of typical curved surfaces. For brain surfaces, we can classify sulcal pits ($-1 < SI < -0.5$), sulcal saddles ($-0.5 < SI < 0$), saddles ($SI = 0$), gyral saddles ($0 < SI < 0.5$) and gyral nodes ($0.5 < SI < 1$). When the shape index equals -1 or 1 , it represents a defect of the curvature tensor with two eigenvalues identical, as shown in Fig. S3. The probability of shape index distribution exhibits two peaks, as shown in **Fig. 4(c)**, corresponding to ridge and rut shapes ($SI = \pm 0.5$), where the ridge shape ($SI = 0.5$) is dominating. In contrast, the rescaled mean curvature histogram exhibits a unique peak around 0.2 (Fig. S3). The two-peak and unique-peak distributions of adult human brain surfaces have also been presented in previous research [36, 37]. To quantify the similarities between every two brain surfaces, we evaluate the distribution of $I(v)$ differences on the common disk domain at each vertex v . Here $I(v)$ represents either the surface shape index $SI(v)$ or the rescaled mean curvature $\tilde{H}(v)$ defined as

$$\tilde{H}(v) = 2 \frac{H(v) - \min(H)}{\max(H) - \min(H)} - 1. \quad (3)$$

Thus the similarity s of the distributions I_{S_1}, I_{S_2} of the two surfaces $\mathcal{S}_1, \mathcal{S}_2$ is then evaluated via

$$s(S_1, S_2) = 1 - \frac{1}{2m^{1/p}} \|I_{S_1} - I_{S_2}(g_2^{-1} \circ f_{12} \circ g_1)\|_p, \quad (4)$$

where m is the total number of vertices, $g_1: \mathcal{S}_1 \rightarrow \mathcal{D}_1$ and $g_2: \mathcal{S}_2 \rightarrow \mathcal{D}_2$ are the initial disk conformal parameterizations, f_{12} is the landmark-aligned quasi-conformal map between $\mathcal{D}_1$ and $\mathcal{D}_2$, and $\|\cdot\|_p$ is the vector p -norm:

$$\|I_{S_1} - I_{S_2}(g_2^{-1} \circ f_{12} \circ g_1)\|_p = \left(\sum_{k=1}^m |I_{S_1}(v_k) - I_{S_2}((g_2^{-1} \circ f_{12} \circ g_1)(v_k))|^p \right)^{1/p}. \quad (5)$$

Note that since $I_{S_1} \in [-1, 1]$ and $I_{S_2} \in [-1, 1]$, we have $0 \leq \|I_{S_1} - I_{S_2}(g_2^{-1} \circ f_{12} \circ g_1)\|_p \leq (2^p m)^{1/p} = 2m^{1/p}$. Therefore, we always have $0 \leq s(\mathcal{S}_1, \mathcal{S}_2) \leq 1$.

We calculate the similarity indices with different p -norm: $p = 1$, $p = 2$, and $p = \infty$ of both rescaled mean curvature and shape index. The results are given in **Table 1** and Table S2.

Discussion

In this study, we have explored the mechanisms underlying brain morphogenesis for a few different mammalian species. Using fetal and adult brain MRIs for ferrets, macaques and humans, we carried out physical experiments using swelling gels, combined with a mathematical framework to model the differential growth of the cortex that leads to iterations and variations of a mechanical (sulcification) instability that allowed us to recapitulate the basic morphological development of folds. We then deployed a range of morphometric tools to compare the results of

Fig. 4

Comparison among real (₁), simulated (₂), and gel brains (₃) of ferret, rhesus macaque, and human via morphometric analysis.

(a) 3D cortical surfaces of *in vivo*, *in silico*, and *in vitro* models. Left brain surfaces are provided here. The symmetry of the left and right halves of brain surfaces is discussed in Fig. S3 and S4, Movie S2–S4. (b) The quasi-conformal disk mapping with landmark matching of cortical surfaces on disk (see Sec *Morphometric analysis* for details). Blue or red curves represent corresponding landmarks. Color represents shape index (SI, as defined in Eq. (2)). Similarity indices of each simulated and gel brain surfaces are presented in Table 1. (c) Histogram of shape index of ferret, macaque, and human. Black, red, and blue dots represent the probability of shape index of real, gel, and simulated surfaces, respectively.

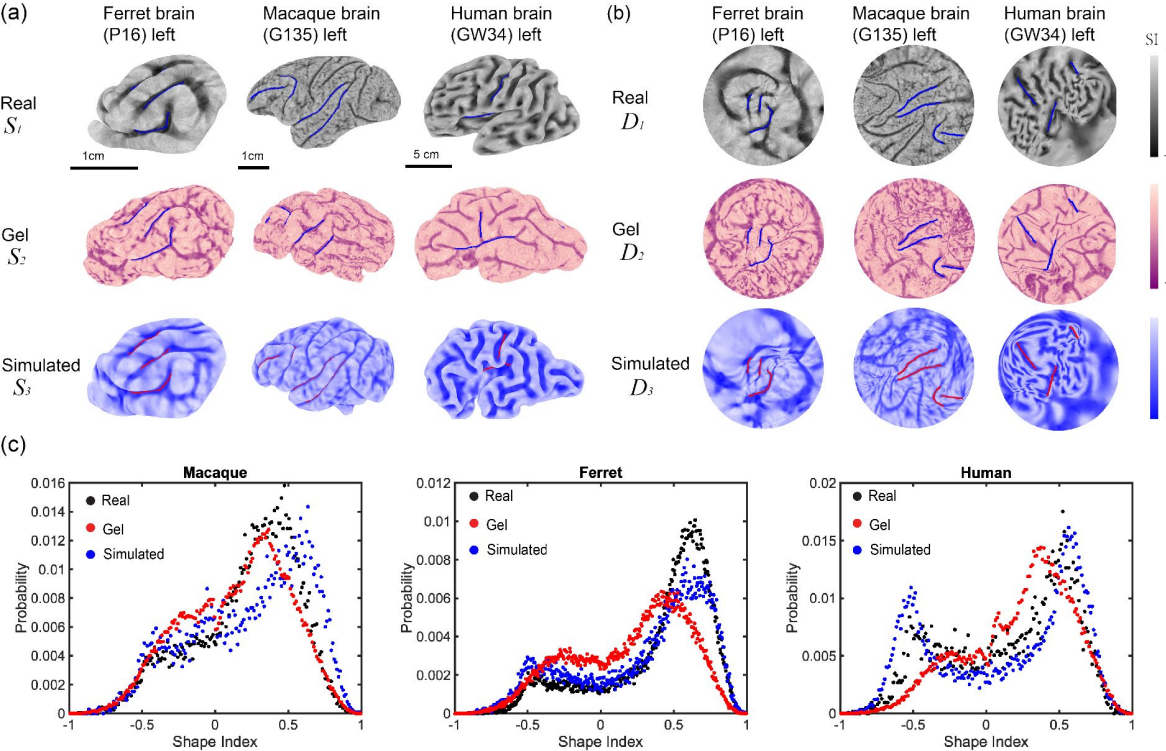


Table 1

Similarity index evaluated by comparing the shape index of simulated brains (S), swollen gel brain simulacrum (G) and real brain surfaces (R), calculated with vector p-norm $p = 2$, as described in Eq. (4).

Similarity Index	simulation-real	gel-real	simulation-gel
Ferret (left)	0.7632	0.7307	0.7117
Ferret (right)	0.7339	0.7222	0.7019
Macaque (left)	0.7624	0.7608	0.7512
Macaque (right)	0.7559	0.7632	0.7463
Human (left)	0.6947	0.7425	0.7000
Human (right)	0.6909	0.7325	0.7001

species	model	g_g/g_w	μ_g/μ_w	h/L
Ferret	step-wise	1.8	1	$0.1 - 0.005t$
Macaque	step-wise	1.8	1	$0.1 - 0.05t, 0.1 - 0.1t$
Human	continuous	1.8	1	$0.05 - 0.03t$

Table 2

Parameters for numerical simulations

our physical experiments and simulations with 3d scans of real brains, and show that our approaches are qualitatively and quantitatively consistent with experimental observations of brain morphologies.

All together, our study shows that differential growth between the gray matter cortex and white matter bulk provides a minimal physical model to explain the variations in the cortical folding patterns seen in multiple species. More specifically, we see that the overall morphologies are controlled by the relative size of the brain (compared to the cortex), as well as the scaled surface expansion rate, both of which can and do have multiple genetic antecedents [38, 39, 40, 41, 42] (see SI, Table S1). Our results point to some open questions. First, the relationship between physical processes that shape organs and the molecular and cellular processes underlying growth has been the subject of many recent studies, e.g. in the context of gut development [43, 44], and it would seem natural to expect similar relationships in brain development. There is a growing literature linking genes with brain malformation and pathologies. For example, GPR56 [38] and Cdk5 [39] can affect progenitors and neurons in migration, SP0535 [40] can affect neural proliferation, and foxp2 can affect neural differentiation [41, 42], all of which change the cortical expansion rate and thickness, consequently leading to brain malformation and pathologies, as listed in Table S1. While a direct relation between gene expression levels and the effective tangential growth rate G and cortical thickness has only been partially resolved, as for example in our companion study on the folding and misfolding of the ferret brain [11], further studies are needed to address how genetic programs drive cell proliferation, migration, size and shape change that ultimately lead to different cortical morphologies. Second, although our focus has been exclusively on the morphology of the brain, recent studies [3, 12] are suggestive of a link between cortical geometry and function from both developmental and evolutionary perspectives, and suggest natural questions for future study. Third, we are not able to set a spatio-temporal distribution of the cortical expansion for fetal brain surfaces of all three species because of a lack of growth rate data *in vivo*. The effect of inhomogeneous growth needs to be further investigated by incorporating regional growth of the gray and white matter not only in human brains [29, 31], but also in other species. Finally, our physical and computational models along with our morphometric approaches are a promising avenue to pursue in the context of the inverse growth problem, i.e., postulate the fetal brain morphologies from the adult brains. This may one day soon allow us to reconstruct the adult brain geometries from fossil endocasts [45], and eventually provide insights into how a few mutations might have triggered the rapid expansion of the cortex across evolutionary time and led to the highly convoluted human brain that is able to ponder the question of how it might have folded itself.

Methods

3D model reconstruction

Pre-processing

We used a publicly available database for all our 3d reconstructions: fetal macaque brain surfaces are obtained from [26]; newborn ferret brain surfaces are obtained from [11]; and fetal human brain surfaces are obtained from [22]. These 2D manifolds of brain surfaces were first normalized by their characteristic lengths $L_0 = L_x^0/2$ and discretized to triangle meshes by Meshlab [46] and then converted to 3D models and discretized to tetrahedral elements in Netgen [47].

Post-processing

All the numerically calculated brain surfaces and scanned gel brain surfaces were extracted as 3D triangle meshes. These intact cortical surfaces were then dissected to left and right semi-brains and normalized by half of their longitudinal lengths $L = L_x/2$. These surfaces were then checked

and fixed to a simply connected open surface to satisfy the requirements for open disk conformal mapping [32].

Experimental protocol for gel experiments

The physical gel was constructed following a previous publication [22]. In brief, a negative rubber mold was generated with Ecoflex 00-30 from a 3D-printed fetal brain plastic model. The core gel was then generated with SYLGARD 184 at a 1:45 crosslinker: base ratio. Three layers of PDMS gel at a 1:35 crosslinker:base ratio were surface-coated onto the core layer to mimic the cortex layer. Pigments were added to the PDMS mixture for bright-field visualization. To mimic the cortex folding process, the physical gel was immersed in n-hexane. The time-lapse videos were taken with an iPod Touch 7th Gen. To reconstruct the swollen gel surface and analyze the folding patterns, swollen gels were imaged with a Nikon X-Tek HMXST X-ray CT machine. The voxel resolution for all scans was 100 μm . To minimize solvent evaporation during the 30-minute scan, cotton soaked in the solvent was placed inside the container. The container's thin acrylic walls allowed for clear X-ray transmission. The container's thin acrylic walls allowed for clear X-ray transmission. To test the reversibility of the folding pattern formation, the physical gel models were allowed to dry in a laminar flow hood overnight before being immersed in hexane again.

3D reconstruction of swollen gel surface from X-ray CT

The z-stack images obtained from the X-ray CT machine were segmented by a machine-learning-based segmentation toolbox, Labkit, via ImageJ. A classification was created for each swollen gel. Then a 3D surface of the segmented gel image was generated by ImageJ 3D viewer. Further post-processing was conducted in SOLIDWORKS and MeshLab, including reversing facial normal direction, re-meshing, and surface-preserving Poisson smoothing.

Numerical Simulations

The simulation geometries of ferret, macaque, and human are based on MRI of smooth fetal or newborn brains. For ferret, we take the P0, P4, P8, and P16 fetal brains as initial shapes of the step-wise growing model; for macaque, we take G80 and G110 fetal brains as initial shapes of the step-wise growing model; and for human, we take GW22 fetal brain as the initial shape of the continuous growing model. Both gray and white matter are considered as neo-Hookean hyperelastic material with the shear modulus distribution

$$\mu(d) = \mu_w + \frac{\mu_g - \mu_w}{1 + \exp[10(d/h - 1)]}, \quad (6)$$

where d is the distance from an arbitrary material point inside the brain to the cortical surface, and h is the approximated cortical thickness assumed to decrease with growth process time t [22]. The subscripts 'g' and 'w' represent gray and white matter, respectively. The tangential growth ratio g has a similar spatial distribution

$$g(d) = g_w + \frac{g_g - g_w}{1 + \exp[10(d/h - 1)]}, \quad (7)$$

where g_w and g_g represent the growth ratio at cortical surface and at innermost white matter. The parameters used in simulations are listed in **Table 2**.

Data availability

Reconstructed 3D surface models of fetal and adult brains of macaque and human are available on GitHub at <https://github.com/YinSifan0204/Comparative-brain-morphologies>. Newborn ferret brain surfaces will be made available upon publication of the paper. All other data are included in the

article and/or supplementary material.

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Additional information

Author contributions

L.M. and R.T. conceived the project; S.Y., C.L., G.P.T.C., and L.M. designed research; S.Y. collected data and performed numerical simulations; C.L. and Y.J. performed gel experiments; S.Y. and G.P.T.C. contributed morphometric analysis; K.H. and R.T. provided adult ferret and primate brain data; S.Y., G.P.T.C., and L.M. wrote the paper with input from all authors; all authors edited the paper; L.M. supervised the project.

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Reviewer #1 (Public review):

The manuscript by Yin and colleagues addresses a long-standing question in the field of cortical morphogenesis, regarding factors that determine differential cortical folding across species and individuals with cortical malformations. The authors present work based on a computational model of cortical folding evaluated alongside a physical model that makes use of gel swelling to investigate the role of a two-layer model for cortical morphogenesis. The study assesses these models against empirically derived cortical surfaces based on MRI data from ferret, macaque monkey, and human brains.

The manuscript is clearly written and presented, and the experimental work (physical gel modeling as well as numerical simulations) and analyses (subsequent morphometric evaluations) are conducted at the highest methodological standards. It constitutes an exemplary use of interdisciplinary approaches for addressing the question of cortical morphogenesis by bringing together well-tuned computational modeling with physical gel models. In addition, the comparative approaches used in this paper establish a foundation for broad-ranging future lines of work that investigate the impact of perturbations or abnormalities during cortical development.

The cross-species approach taken in this study is a major strength of the work. However, correspondence across the two methodologies did not appear to be equally consistent in predicting brain folding across all three species. The results presented in Figures 4 (and Figures S3 & S4) show broad correspondence in shape index and major sulci landmarks across all three species. Nevertheless, the results presented for the human brain lack the same degree of clear correspondence for the gel model results as observed in the macaque and ferret. While this study clearly establishes a strong foundation for comparative cortical anatomy across species and the impact of perturbations on individual morphogenesis, further work that fine-tunes physical modeling of complex morphologies, such as that of the human cortex, may help to further understand the factors that determine cortical functionalization and pathologies.

<https://doi.org/10.7554/eLife.107138.1.sa2>

Reviewer #2 (Public review):

This manuscript explores the mechanisms underlying cerebral cortical folding using a combination of physical modelling, computational simulations, and geometric morphometrics. The authors extend their prior work on human brain development (Tallinen et al., 2014; 2016) to a comparative framework involving three mammalian species: ferrets (Carnivora), macaques (Old World monkeys), and humans (Hominoidea). By integrating swelling gel experiments with mathematical differential growth models, they simulate sulcification instability and recapitulate key features of brain folding across species. The authors make commendable use of publicly available datasets to construct 3D models of fetal

and neonatal brain surfaces: fetal macaque (ref. [26]), newborn ferret (ref. [11]), and fetal human (ref. [22]).

Using a combination of physical models and numerical simulations, the authors compare the resulting folding morphologies to real brain surfaces using morphometric analysis. Their results show qualitative and quantitative concordance with observed cortical folding patterns, supporting the view that differential tangential growth of the cortex relative to the subcortical substrate is sufficient to account for much of the diversity in cortical folding. This is a very important point in our field, and can be used in the teaching of medical students.

Brain folding remains a topic of ongoing debate. While some regard it as a critical specialization linked to higher cognitive function, others consider it an epiphenomenon of expansion and constrained geometry. This divergence was evident in discussions during the Strüngmann Forum on cortical development (Silver et al., 2019). Though folding abnormalities are reliable indicators of disrupted neurodevelopmental processes (e.g., neurogenesis, migration), their relationship to functional architecture remains unclear. Recent evidence suggests that the absolute number of neurons varies significantly with position-sulcus versus gyrus-with potential implications for local processing capacity (e.g., <https://doi.org/10.1002/cne.25626>). The field is thus in need of comparative, mechanistic studies like the present one.

This paper offers an elegant and timely contribution by combining gel-based morphogenesis, numerical modelling, and morphometric analysis to examine cortical folding across species. The experimental design - constructing two-layer PDMS models from 3D MRI data and immersing them in organic solvents to induce differential swelling - is well-established in prior literature. The authors further complement this with a continuum mechanics model simulating folding as a result of differential growth, as well as a comparative analysis of surface morphologies derived from in vivo, in vitro, and in silico brains.

I offer a few suggestions here for clarification and further exploration:

Major Comments

(1) Choice of Developmental Stages and Initial Conditions

The authors should provide a clearer justification for the specific developmental stages chosen (e.g., G85 for macaque, GW23 for human). How sensitive are the resulting folding patterns to the initial surface geometry of the gel models? Given that folding is a nonlinear process, early geometric perturbations may propagate into divergent morphologies. Exploring this sensitivity-either through simulations or reference to prior work-would enhance the robustness of the findings.

(2) Parameter Space and Breakdown Points

The numerical model assumes homogeneous growth profiles and simplifies several aspects of cortical mechanics. Parameters such as cortical thickness, modulus ratios, and growth ratios are described in Table II. It would be informative to discuss the range of parameter values for which the model remains valid, and under what conditions the physical and computational models diverge. This would help delineate the boundaries of the current modelling framework and indicate directions for refinement.

(3) Neglected Regional Features: The Occipital Pole of the Macaque

One conspicuous omission is the lack of attention to the occipital pole of the macaque, which is known to remain smooth even at later gestational stages and has an unusually high neuronal density (2.5× higher than adjacent cortex). This feature is not reproduced in the gel or numerical models, nor is it discussed. Acknowledging this discrepancy-and speculating on

possible developmental or mechanical explanations-would add depth to the comparative analysis. The authors may wish to include this as a limitation or a target for future work.

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The authors suggest that their morphometric framework could be extended to solve the inverse growth problem-reconstructing fetal geometries from adult brains. This speculative but intriguing direction has implications for evolutionary neuroscience, particularly the interpretation of fossil endocasts. Although beyond the scope of this paper, I encourage the authors to elaborate briefly on how such a framework might be practically implemented and validated.

Conclusion

This is a well-executed and creative study that integrates diverse methodologies to address a longstanding question in developmental neurobiology. While a few aspects-such as regional folding peculiarities, sensitivity to initial conditions, and available human data-could be further elaborated, they do not detract from the overall quality and novelty of the work. I enthusiastically support this paper and believe that it will be of broad interest to the neuroscience, biomechanics, and developmental biology communities.

Note: The paper mentions a companion paper [reference 11] that explores the cellular and anatomical changes in the ferret cortex. I did not have access to this manuscript, but judging from the title, this paper might further strengthen the conclusions.

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Author response:

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The manuscript by Yin and colleagues addresses a long-standing question in the field of cortical morphogenesis, regarding factors that determine differential cortical folding across species and individuals with cortical malformations. The authors present work based on a computational model of cortical folding evaluated alongside a physical model that makes use of gel swelling to investigate the role of a two-layer model for cortical morphogenesis. The study assesses these models against empirically derived cortical surfaces based on MRI data from ferret, macaque monkey, and human brains.

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We thank the reviewer for positive opinions and helpful comments. Yes, the physical gel model of the human brain has a lower similarity index with the real brain. There are several reasons.

First, the highly convoluted human cortex has a few major folds (primary sulci) and a very large number of minor folds associated with secondary or tertiary sulci (on scales of order comparable to the cortical thickness), relative to the ferret and macaque cerebral cortex. In our gel model, the exact shapes, positions, and orientations of these minor folds are stochastic, which makes it hard to have a very high similarity index of the gel models when compared with the brain of a single individual.

Second, in real human brains, these minor folds evolve dynamically with age and show differences among individuals. In experiments with the gel brain, multiscale folds form and eventually disappear as the swelling progresses through the thickness. Our physical model results are snapshots during this dynamical process, which makes it hard to have a concrete one-to-one correspondence between the instantaneous shapes of the swelling gel and the growing human brain.

Third, the growth of the brain cortex is inhomogeneous in space and varying with time, whereas, in the gel model, swelling is relatively homogeneous.

We agree that further systematic work, based on our proposed methods, with more fine-tuned gel geometries and properties, might provide a deeper understanding of the relations between brain geometry, and growth-induced folds and their functionalization and pathologies. Further analysis of cortical pathologies using computational and physical gel models can be found in our companion paper (Choi et al., 2025), also submitted to eLife:

G. P. T. Choi, C. Liu, S. Yin, G. Sejourne, R. S. Smith, C. A. Walsh, L. Mahadevan, Biophysical basis for brain folding and misfolding patterns in ferrets and humans. Preprint, bioRxiv 2025.03.05.641682.

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We thank the reviewer for the very positive comments.

I offer a few suggestions here for clarification and further exploration:

Major Comments

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The authors should provide a clearer justification for the specific developmental stages chosen (e.g., G85 for macaque, GW23 for human). How sensitive are the resulting folding patterns to the initial surface geometry of the gel models? Given that folding is a nonlinear process, early geometric perturbations may propagate into divergent morphologies. Exploring this sensitivity-either through simulations or reference to prior work-would enhance the robustness of the findings.

The initial geometry is one of the important factors that decides the final folding pattern. The smooth brain in the early developmental stage shows a broad consistency across individuals, and we expect the main folds to form similarly across species and individuals.

Generally, we choose the initial geometry when the brain cortex is still relatively smooth. For the human, this corresponds approximately to GW23, as the major folds such as the Rolandic fissure (central sulcus), arise during this developmental stage. For the macaque brain, we chose developmental stage G85, primarily because of the availability of the dataset corresponding to this time, which also corresponds to the least folded.

We expect that large-scale folding patterns are strongly sensitive to the initial geometry but fine-scale features are not. Since our goal is to explain the large-scale features, we expect sensitivity to the initial shape.

Enclosed are some results from other researchers that are consistent with this idea. Below are some images of simulations from Wang et al. obtained by perturbing the geometry of a sphere to an ellipsoid. We see that the growth-induced folds mostly maintain their width (wavelength), but change their orientations.

Reference:

Wang, X., Lefevre, J., Bohi, A., Harrach, M.A., Dinomais, M. and Rousseau, F., 2021. *The influence of biophysical parameters in a biomechanical model of cortical folding patterns*. *Scientific Reports*, 11(1), p.7686.

Related results from the same group show that slight perturbations of brain geometry, cause these folds also tend to change their orientations but not width/wavelength (Bohi et al., 2019).

Reference:

Bohi, A., Wang, X., Harrach, M., Dinomais, M., Rousseau, F. and Lefevre, J., 2019, July. *Global perturbation of initial geometry in a biomechanical model of cortical morphogenesis*. In *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (pp. 442-445). IEEE.

Finally, a systematic discussion of the role of perturbations on the initial geometries and physical properties can be seen in our work on understanding a different system, gut morphogenesis (Gill et al., 2024).

We have added the discussion about geometric sensitivity in the section Methods-Numerical Simulations:

“Small perturbations on initial geometry would affect minor folds, but the main features of major folds, such as orientations, width, and depth, are expected to be conserved across individuals [49, 50]. For simplicity, we do not perturb the fetal brain geometry obtained from datasets.”

(2) Parameter Space and Breakdown Points

The numerical model assumes homogeneous growth profiles and simplifies several aspects of cortical mechanics. Parameters such as cortical thickness, modulus ratios, and growth ratios are described in Table II. It would be informative to discuss the range of parameter values for which the model remains valid, and under what conditions the physical and computational models diverge. This would help delineate the boundaries of the current modelling framework and indicate directions for refinement.

Exploring the valid parameter space is a key problem. We have tested a series of growth parameters and will state them explicitly in our revision. In the current version, we chose the ones that yield a relatively high similarity index to the animal brains. More generally, folding patterns are largely regulated by geometry as well as physical parameters, such as cortical thickness, modulus ratios, growth ratios, and inhomogeneity. In our previous work on a different system, gut morphogenesis, where similar folding patterns are seen, we have explored these features (Gill et al., 2024).

Reference:

Gill, H.K., Yin, S., Nerurkar, N.L., Lawlor, J.C., Lee, C., Huycke, T.R., Mahadevan, L. and Tabin, C.J., 2024. Hox gene activity directs physical forces to differentially shape chick small and large intestinal epithelia. *Developmental Cell*, 59(21), pp.2834-2849.

(3) Neglected Regional Features: The Occipital Pole of the Macaque

One conspicuous omission is the lack of attention to the occipital pole of the macaque, which is known to remain smooth even at later gestational stages and has an unusually high neuronal density (2.5× higher than adjacent cortex). This feature is not reproduced in the gel or numerical models, nor is it discussed. Acknowledging this discrepancy and speculating on possible developmental or mechanical explanations would add depth to the comparative analysis. The authors may wish to include this as a limitation or a target for future work.

Yes, we have added that the omission of the Occipital Pole of the macaque is one of our paper's limitations. Our main aim in this paper is to explore the formation of large-scale folds, so the smooth region is neglected. But future work could include this to make the model more complete.

The main text has been modified in Methods, 3D model reconstruction, pre-processing:

“To focus on fold formation, we neglected some smooth regions such as the Occipital Pole of the macaque.”

(4) Spatio-Temporal Growth Rates and Available Human Data

The authors note that accurate, species-specific spatio-temporal growth data are lacking, limiting the ability to model inhomogeneous cortical expansion. While this may be true for ferret and macaque, there are high-quality datasets available for human fetal development, now extended through ultrasound imaging (e.g., <https://doi.org/10.1038/s41586-023-06630-3>). Incorporating or at least referencing such data could improve the fidelity of the human model and expand the applicability of the approach to clinical or pathological scenarios.

We thank the reviewer for pointing out the very useful datasets that exist for the exploration of inhomogeneous growth driven folding patterns. We have referred to this paper to provide suggestions for further work in exploring the role of growth inhomogeneities.

We have referred to this high-quality dataset in our main text, Discussion:

“...the effect of inhomogeneous growth needs to be further investigated by incorporating regional growth of the gray and white matter not only in human brains [29, 31] based on public datasets [45], but also in other species.”

A few works have tried to incorporate inhomogeneous growth in simulating human brain folding by separating the central sulcus area into several lobes (e.g., lobe parcellation method, Wang, PhD Thesis, 2021). Since our goal in this paper is to explain the large-scale features of folding in a minimal setting, we have kept our model simple and show that it is still capable of capturing the main features of folding in a range of mammalian brains.

Reference:

Xiaoyu Wang. Modelisation et caractérisation du plissement cortical. Signal and Image Processing. Ecole nationale supérieure Mines-Télécom Atlantique, 2021. English. (NNT : 2021IMTA0248).

(5) Future Applications: The Inverse Problem and Fossil Brains

The authors suggest that their morphometric framework could be extended to solve the inverse growth problem-reconstructing fetal geometries from adult brains. This

speculative but intriguing direction has implications for evolutionary neuroscience, particularly the interpretation of fossil endocasts. Although beyond the scope of this paper, I encourage the authors to elaborate briefly on how such a framework might be practically implemented and validated.

For the inverse problem, we could use the following strategies:

- a. Perform systematic simulations using different geometries and physical parameters to obtain the variation in morphologies as a function of parameters.
- b. Using either supervised training or unsupervised training (physics-informed neural networks, PINNs) to learn these characteristic morphologies and classify their dependence on the parameters using neural networks. These can then be trained to determine the possible range of geometrical and physical parameters that yield buckled patterns seen in the systematic simulations.
- c. Reconstruct the 3D surface from fossil endocasts. Using the well-trained neural network, it should be possible to predict the initial shape of the smooth brain cortex, growth profile, and stiffness ratio of the gray and white matter.

As an example in this direction, supervised neural networks have been used recently to solve the forward problem to predict the buckling pattern of a growing two-layer system (Chavoshnejad et al., 2023). The inverse problem can then be solved using machine-learning methods when the training datasets are the folded shape, which are then used to predict the initial geometry and physical properties.

Reference:

Chavoshnejad, P., Chen, L., Yu, X., Hou, J., Filla, N., Zhu, D., Liu, T., Li, G., Razavi, M.J. and Wang, X., 2023. An integrated finite element method and machine learning algorithm for brain morphology prediction. *Cerebral Cortex*, 33(15), pp.9354-9366.

Conclusion

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